William Dennis

PhD Researcher (PrO-AI CDT), University of Bristol
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Research interests: robust and reliable machine learning; evaluation and benchmarking; applied reinforcement learning; healthcare and wearables.

Experience

Honorary Research Associate in Machine Learning — University of Bristol Oct 2024 – Present
Bristol, UK

- Released and active owner of an open-source fault-injection framework used within the department for the wider research community.
- Actively exchanged knowledge with industry stakeholders on fault-tolerance methods for AI systems.

Research Intern — University of Bristol x National University of Singapore Jun 2026 – Jul 2026
Bristol

- Researched reinforcement learning methods for health behaviour change, supervised by Dr. Mengyan Zhang (University of Bristol) and Swapnil Mishra (National University of Singapore).
- Sole developer of rl-health-interventions: a config-first simulator for Just-In-Time Adaptive Interventions.

Energy Modeller — Aurora Energy Research Oct 2024 – Jun 2026
Oxford, UK

- Forecasted the impact of AI datacentre demand and collaborated with clients interested in SMRs.
- Applied modelling methods using convex optimisation, linear programming, and maximum likelihood estimation with a core focus on explainability, robustness and interpretability.
- Researched and presented a business case for reinforcement learning to improve energy pricing.
- Led modelling of the Polish, Czech and Slovak energy market models, tailoring foundational models to regional needs and building internal toolkits across Europe with Streamlit and SQL.
- Applied best software engineering practices, including test-driven development, Git version control, and agile/scrum workflows.

Data Scientist — JPMorgan Feb 2024 – Jul 2024
Bristol, UK

- Led a team of 4 data scientists to produce in-house quant level research into the development of trading signals derived from millisecond level limit-order-book data.
- Built tooling for my team to parse millions of rows of raw text data on local machines using JIT.
- Created baseline models using ARIMA, RNNs and LSTMs on level-1 data.
- Engineered 50+ features derived from cleaned up level-2 limit order book data.

Signal Processing Engineer — Milbotix Feb 2024 – Oct 2024
Bristol Robotics Lab, UK

- Developed a two-stage machine learning pipeline optimised for use on embedded devices using accelerometer filtering and novel Fourier features.
- Reduced pipeline latency from minutes to seconds via an accelerometer-based filtering layer and JIT compilation.
- Achieved an error rate of <3bpm, allowing the device to be medical grade.
- Extended to lead validation for medical-grade labelling.

Intern Hedging Analyst (Risk Team) — Ocean Partners Jul 2021 – Jul 2022
Maidenhead, UK

- Mitigated £250m of physical commodity transactions via the London Metal Exchange.
- Maintained a strong relationship with banks and brokers using responsive and professional communication.
- Automated a data analysis task for clearing transactions, saving 60 minutes of manual labour daily.

Education

University of Bristol — PhD, PrO-AI CDT Sep 2026 – Present

Doctoral Researcher

- Funded doctoral place under Professor Peter Flach, Department of Computer Science.

University of Bristol — MSc Data Science Sep 2023 – Sep 2024

Distinction (Top 3 of cohort)

- Researched theory of neural network generalisation: grokking, double descent, Lipschitz continuity.
- Deployed distributed cloud architectures on AWS (DynamoDB, SQS, SNS, S3, EC2).
- Studied technology ethics, legal awareness (GDPR), NLP, and the history of technological innovation.
- Produced a Tableau report analysing migration patterns of UK residents aged 20-25.
- Active member of Google Developer Student Club; Quantitative Analyst, Bristol Trading Society.

University of Bath — BSc (Hons) Mathematical Sciences Sep 2019 – Jul 2023

Scientific Computing (84%) and Machine Learning (91%)

- Built websites in Python to support startups in SETsquared using Django.
- 2019/2020 poker society winner.

Research

Robust ML in Harsh Environments 2025

A novel framework in Python with PyTorch for evaluating neural networks in harsh environments. Proposed two novel activation functions that proved more robust to SEE-related errors.

Framework (GitHub) | Interactive Explanation

Reinforcement Learning for Health Behaviour Change 2026

Config-first simulation environment for Just-in-Time Adaptive Interventions (JITAIs) with bandit and RL baselines, heterogeneous user models, and an LLM user simulator.

GitHub

SLM Bias Testing 2026

Benchmarking framework for bias in small language models, covering StereoSet, WinoBias, CV screening, and demographic output-length disparity. Evaluates how bias changes across model size, family, and release date.

GitHub

Classification of Signal Quality from Ankle-Based PPG Signals 2024–2025

Research project with Milbotix developing novel time-series features and neural-network classifiers for ankle-based PPG signal-quality classification, evaluated with AUC and ROC metrics.

Publication

Publications

1. **William Dennis** and James Pope. *Robust Machine Learning for Harsh Environments: A Framework and Evaluation of Key Architectural Choices.* Lecture Notes in Computer Science, Vol. 16517 (LNAI), Jun 2026. Published. https://doi.org/10.1007/978-3-032-25032-2_10
2. **William Dennis** and James Pope. *A Framework for Developing Robust Machine Learning Models in Harsh Environments: A Review of CNN Design Choices.* ICAART 2025 (17th International Conference on Agents and Artificial Intelligence), Feb 2025. Published. https://research-information.bris.ac.uk/en/publications/a-framework-for-developing-robust-machine-learning-models-in-hars/
3. Steer, Z., Venkatesh, P. T., Mejia-Mejia, E., **Dennis, W. W.**, Ogundele, P. A., Brooking, A., & Eimontaite, I. *SmartSocks: a new data collection paradigm for dementia and other neurological disorders.* Alzheimer's & Dementia, 20(S8), e095065, 2025. Published. https://doi.org/10.1002/alz.095065

Technical Skills

- **Languages:** Python (10+ years), SQL, C#, R, MATLAB, Haskell, TypeScript, GAMS
- **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, Streamlit, Django
- **Cloud & Tools:** AWS, Google Cloud, Docker, Git, JIRA, Agile

Selected Technical Projects

Moonboard-Analysis 2025-2026

ML benchmarking framework for climbing route prediction. Designed a reproducible evaluation protocol with 5-fold retrain-per-fold cross-validation, tested 8+ model architectures (MLP, LSTM, CNN, Transformer, Ridge, Perceptron), and established a non-tree leaderboard. Built dataset augmentation pipeline with hold-swap tokenisation and identified/patched data leakage bugs.

GitHub | Benchmark Results

OpenFrontBench 2026

Open-source RTS agent benchmark built on a pinned OpenFrontIO engine, with scripted and MCP-based agents, replay verification, and analytical policy baselines.

GitHub

RL Trader 2023

Reinforcement-learning trading agent using linear features.

Activities

- Presented at ICAART 2025 — 17th International Conference on Agents and Artificial Intelligence, Feb 2025.
- Attended Pre-ICML London 2025 — UCL Bloomsbury Campus, Jul 2025.
- Attended AE Global Summit on Open Problems for AI — Friends House, London, Oct 2025.
- Attended UK Theoretical Neuroscience Workshop, Dec 2025.
- Attended BAIS Summer School 2026 — Engineers' House, Bristol, Sep 2026.